Cover Letter

July 4, 2026

Editorial Office

Data & Knowledge Engineering

Elsevier

Dear Editor,

We are pleased to submit our manuscript, "Structurally-Calibrated Functional Attribution for Audit Prioritization in Knowledge-Intensive Reasoning", for consideration as a regular article in Data & Knowledge Engineering.

The manuscript presents Structurally-Calibrated Functional Attribution (SC-FMA), a graph-aware method for converting structured intermediate knowledge objects into auditable priority records. Its novelty is to treat functional attribution as a representation for knowledge audit: local utility or annotation-fidelity signals are preserved while structural dependency, redundancy, and bottleneck roles are recorded as curation fields.

The manuscript fits Data & Knowledge Engineering because it addresses structured audit prioritization for knowledge-intensive workflows, aligning with the journal's interests in knowledge representation, knowledge-base maintenance, and management of engineered data and knowledge artifacts. SC-FMA turns traces of retrieval checks, entity bindings, rule-like steps, and process annotations into graph-aware audit records for fixed-budget curation.

The evidence package includes locked PRM800K annotation evidence where w_struct is the primary real-data signal and Ridge preserves it while exposing a decomposable audit record, a Countries-KG representation study showing that typed ontology edges can change structural-priority rankings, rule-derived audit-target retrieval, and controlled synthetic calibration. The manuscript keeps deployment, downstream-training, external-transfer, and causal claims separate from the reported audit-prioritization evidence.

This work was supported by the National Social Science Fund of China Project (24BSH018) and the Beijing Natural Science Foundation Project (L252145). The authors declare that they have no conflicts of interest related to this work. The manuscript's generative-AI disclosure states that OpenAI GPT-5 was used to improve language clarity and assist with LaTeX formatting checks, with all content reviewed by the authors. PRM800K, MuSiQue, and WebQSP are publicly available from their original sources; derived reports and reproduction artifacts will be deposited in an anonymous public repository for review and released with the final article.

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Sincerely,

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